Supplementary Material for RAPTOR: Resource-Aware Physics-Token Operator with Residual Local Adaptation for Aerodynamic Field Reconstruction

Illustrative AirfRANS pressure-error cases

Figure S1 preserves the two seed-42 pressure-divided-by-density (p/ρ) examples formerly shown in the main manuscript. The upper row is from Reynolds-number extrapolation and compares RAPTOR with Transolver-Tiny; the lower row is from angle-of-attack (AoA) extrapolation and compares RAPTOR with LinearNO-Tiny. The comparator in each row ranks second by the four-variable macro relative L_2 error in the corresponding seed-42 experiment.

Each case contains $N = 4096$ sampled nodes. For model a, test case q, and the p/ρ channel, define the sample-level error and pooled test-set error as

$$e_{q,p}^{(a)} = \frac{\|\widehat{\mathbf{Y}}_{q,p}^{(a)} - \mathbf{Y}_{q,p}\|_2}{\|\mathbf{Y}_{q,p}\|_2 + 10^{-12}}, \qquad \varepsilon_p^{(a)} = \frac{\sqrt{\sum_{q=1}^{M} \|\widehat{\mathbf{Y}}_{q,p}^{(a)} - \mathbf{Y}_{q,p}\|_2^2}}{\sqrt{\sum_{q=1}^{M} \|\mathbf{Y}_{q,p}\|_2^2} + 10^{-12}}. \tag{S1}$$

Here $M = 200$ for Reynolds-number extrapolation and $M = 196$ for AoA extrapolation. Let R denote RAPTOR and b the comparator in the same row. The displayed test case minimizes the distance between its paired pressure improvement and the pooled test-set pressure improvement:

$$\Delta_p^{\text{test}} = 1 - \frac{\varepsilon_p^{(R)}}{\varepsilon_p^{(b)}}, \qquad \Delta_{q,p} = 1 - \frac{e_{q,p}^{(R)}}{e_{q,p}^{(b)}}, \qquad q^* = \arg\min_q \left|\Delta_{q,p} - \Delta_p^{\text{test}}\right|. \tag{S2}$$

The selected test indices are 165 and 142 for Reynolds-number and AoA extrapolation, respectively. The pooled/selected improvements are 4.84%/4.97% and 4.63%/4.17%, respectively. On these selected cases, RAPTOR changes the sample-level p/ρ relative L_2 error from 0.277 to 0.263 and from 0.274 to 0.262, respectively. All metrics use the sampled nodes; percentile normalization affects only the displayed absolute-error colors.

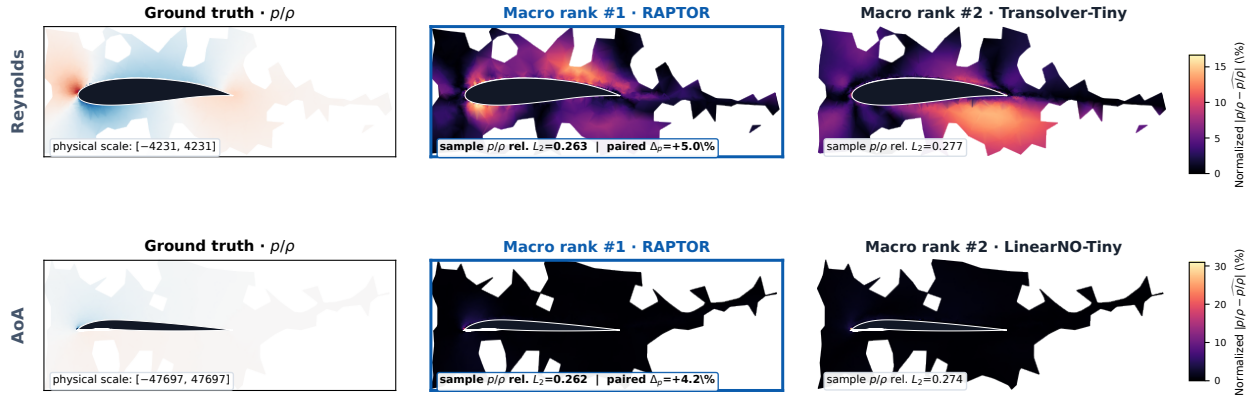


Figure S1: Seed-42 AirfRANS p/ρ ground truth and normalized absolute-error maps. Reynolds-number test index 165 (upper row) compares RAPTOR with Transolver-Tiny; AoA test index 142 (lower row) compares RAPTOR with LinearNO-Tiny. The cases are selected by Equation (S2), not by the largest visual or sample-level improvement. These seed-42 maps are illustrative; they do not establish cross-seed stability or a systematic physical correction mechanism.